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Variable flow control device, system, and method

Abstract

A device, system and method are provided for controlling the rate of infusion of fluids during infusion therapy using non-electric infusion devices. Rotation of a flow regulator dial causes an orifice connected to the inlet to modify its position relative to a particular one or more orifices or groove portions, the characteristics of which provide a certain flow rate characteristic. The flow regulator allows for the infusion pump to infuse at a rate that may be varied during use by the user. Additionally, the flow regulator is made from a material selected to operate under a wide range of pressures, from 5-40 PSI, making the flow regulator compatible with pressurized devices, such as, a non-electric pump.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/879,702 filed on Aug. 2, 2022 which, in turn, is a continuation of U.S. patent application Ser. No. 16/933,946 filed on Jul. 20, 2020 which, in turn, is a divisional of U.S. patent application Ser. No. 15/709,759, filed Sep. 20, 2017; which is a continuation of U.S. patent application Ser. No. 15/397,153, filed Jan. 3, 2017; which is a continuation of U.S. patent application Ser. No. 13/690,702, filed Nov. 30, 2012, which claims the benefit of U.S. Patent Provisional Application No. 61/565,120, filed Nov. 30, 2011; the contents of each of these applications are incorporated herein by reference in their entirety.

BACKGROUND OF THE INVENTION
Field of the Invention
(1) The present invention relates to a device, system and method useful in infusion therapy, and more particularly, useful for varying the flow rate during infusion therapy.
Description of the Related Art
(2) Infusion therapy requires the use of an infusion device (a source of positive pressure). There are several types of infusion devices which include: mechanical pumps, elastomeric pumps, gravity flow, electric/electronic pumps among others. Non-electric pumps and gravity infusions have a

general disadvantage in that they often do not provide a sufficiently stable flow rate.
(3) Flow rate control in mechanical, elastomeric and other non-electrical pumps is generally accomplished with the use of certain small diameter tubing (rate set) that regulates the flow. This presents the following limitations:
(4) The flow cannot be adjusted during the infusion. Instead a new infusion set has to be used when a different rate is required. This adds cost and it may increase the risk of contamination.
(5) In order to change the flow rate, the tubing diameter has to change and thus multiple rate sets have to be made available and changed during infusion. This may or may not be possible during certain therapies.
(6) The nominal flow rate of these sets does not correspond to the flow rate during use due to the viscosity of the fluid often leading to patient and clinician confusion and errors.
(7) Flow rate control in gravity infusions is generally accomplished with roller clamps or flow regulators that allow clinicians to determine a certain position to obtain a desired flow rate. Roller clamps are imprecise and they have generally no flow rate markings. Flow regulators in the prior art offer limited accuracy, versatility and pressure rating performance.
(8) A clinician using flow regulators is generally unaware of the various factors that affect the performance of flow regulators including, the imprecise position of the flow regulator, relative temperature, relative humidity, patient backpressure factors, and the variability of pressure from the source of the medication. These factors can result in significant variances in flow rates and could adversely affect patients to a significant extent.
(9) Flow rate controllers are generally labeled in ml/hour without taking into account the specific effect of the viscosity of the fluid which has a significant effect on the flow rate thus invalidating the significance of the markings of the device and confusing the clinician.
(10) Safety concerns regarding infusions have been escalating in hospitals and in regulatory circles. The FDA has started presenting new guidance documents that regulate infusion system submissions to increase the threshold of requirements for such infusion systems.
(11) Additional design requirements are becoming more apparent in Europe, Canada, Japan, the US and many other countries relative to improved control of flow rates and specific material biocompatibility regulations for fluid delivery devices.
(12) Non-electric infusions systems are generally controlled by certain small diameter tubing (rate set) that regulates the flow. This method presents limitations including inability to change flow rate without changing the rate set, incorrect flow rate labeling due to the varying viscosities of fluids administered, and undesired flow rates due to device design limitations, patient and environmental factors. U.S. Pat. No. 4,904,239 (“the '239 patent”) to Winchell et al., discloses an infusor having a distal flow regulator for dispensing a liquid under pressure at a predetermined flow rate. The '239 patent discloses the use of a non-adjustable, preselected flow regulator including a capillary bore. Col. 5 of the '239 patent, lines 9-14, disclose that a seal design permits the use of dramatically different length regulators for different desired flow rates, while still using the same size housing and connecting means, i. e, the preselected flow rate of the infusor can be changed simply by changing the length of the flow regulator. Thus, a particular flow regulator of the '239 patent has limited flow control characteristics.
(13) U.S. Pat. No. 5,009,251 (“the '251 patent”) to Pike et al., discloses a variable fluid flow controller for regulating the rate of flow from a source of fluid under pressure, including a plurality of unique flow restriction passageways, a valve associated with each passageway and a rotatable cam for selectively opening any one of the valves while maintaining the remaining valves closed. The flow restriction passageway of the '251 patent preferably comprises a channel etched on the surface of a first silicon wafer and enclosed by a second wafer to form a fluid flow passageway, one of the first or second wafers having a plurality of apertures therethrough for intersecting the passageway at various distances along its length.
(14) U.S. Pat. No. 5,234,413 (“the '413 patent”) to Wonder et al., discloses an infusion rate

regulating device for varying the rate of flow of fluids for infusion to a patient at extremely low, but constant, flow rates. The regulator of the '413 patent is interposed at a point on a supply tube between a fluid reservoir and a patient. An input port directs fluid to a fluid metering groove of variable cross-sectional area on a metering plate which is formed as a part of the output port. The metering plate is rotated axially, relative to the input port, allowing fluid to enter the fluid metering groove at any point and flow toward the output port through a fluid metering groove which increases in depth or cross-sectional area at an essentially constant rate. Depending on the point at which the fluid enters the fluid metering groove flow path of the device in the '413 patent, the flow rate selected can be any rate from full off to full flow. Typically elastomeric devices operate in ranges under 5 psi.

(15) What is needed is a flow control device for a gravity flow or mechanical infusion system that provides clinicians with precision in controlling the flow rate through the device for ranges of pressure higher than conventional elastomeric devices.

SUMMARY OF THE INVENTION

(16) A device, system and method are provided for controlling the rate of infusion of fluids during infusion therapy using non-electric infusion devices. The flow control device of the present invention improves flow control, as well as safety resulting from such improved flow rate control, when compared to the performance of flow regulator devices in the prior art. The flow control device of the present invention has a design and method of construction that optimizes flow rate and functionality, safety and ergonomics in applications such as those that can be used with non-electric pumps including, but not limited to: mechanical pumps, elastomeric pumps, and other similar devices or applications.

(17) An embodiment of the disclosure is a flow rate control device configured for connection in a flow path between a non-electric infusion pump and a patient, the flow rate control device comprising: an inlet handle; an outlet handle; a fluid path disposed between said inlet handle and said outlet handle, said fluid path having manually adjustable dimensions; and said inlet handle and said outlet handle composed of materials that can withstand pressures of from 5 PSI-40 PSI. In an embodiment, the flow rate control device further comprises a plurality of differently sized orifices for fluid flow therethrough located on at least one of said inlet handle or said outlet handle. In an embodiment, the flow rate control device further comprises a seal sealingly engaged between said inlet handle and said outlet handle.

(18) An embodiment of the disclosure is a variable flow rate infusion system, comprising: a non-electric infusion pump that dispenses fluid pressurized to between 5 PSI and 40 PSI; and a flow control device according to claim **1** in fluid communication with said non-electric infusion pump. In an embodiment, said materials include at least one of polycarbonate and another material having a similar hardness coefficient to polycarbonate.

(19) In one particular embodiment of the invention, a variable flow control device is provide in which the rotation of a flow regulator dial causes an orifice connected to the inlet to modify its relative position with respect to a groove or open-topped channel connected to the fluid outlet, via an orifice at one end of the groove, thus defining the fluid path. In this embodiment, one or more characteristics (diameter, width, depth, etc.) of the groove may be varied along the length of the groove, as desired.

(20) In another particular embodiment of the invention, rotation of a regulator dial causes an orifice connected to the inlet to align with one of a plurality of orifices connected to the outlet. The diameter of each orifice of the plurality may be graduated such that the different orifices represent a percentage of flow from 1% to 100% in specific increments. In one particular embodiment, ten orifices are provided each orifice providing a 10% greater flow rate of the total possible flow rate than its immediately prior neighbor, starting from the smallest orifice to the largest, with the first orifice providing 10% of the total possible flow rate and the tenth orifice providing 100% of the total possible flow rate.

(21) In a further particular embodiment of the invention, rotation of a flow regulator dial causes an orifice connected to the inlet to align with one of various combinations of orifices connected to the outlet that represent the permutation of orifices as a digital counter (BINARY).
(22) Additionally, in a further particular embodiment of the invention, improved flow regulation or control is achieved by combining two or more adjustable dial layers, each having a variable flow control mechanism in accordance with the present invention. In one particular embodiment, one or more additional variable flow control layers are added after the first or main variable flow control layer to provide a combination of coarse and fine control levels, thus greatly enhancing the actual flow rate controllability through the device.
(23) The present invention solves important limitations inherent to non-electric infusion systems. Other features which are considered as characteristic for the invention are set forth in the drawings and the appended claim.
(24) Although the invention is illustrated and described herein as embodied in a variable flow control device, system and method, it is nevertheless not intended to be limited to the details shown, since various modifications and structural changes may be made therein without departing from the spirit of the invention and within the scope and range of equivalents of the claims.
(25) The construction of the invention, however, together with additional objects and advantages thereof will be best understood from the following description of the specific embodiment when read in connection with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) For a fuller understanding of the nature of the present invention, reference should be made to the following detailed description, taken in connection with the accompanying drawings in which like reference numerals refer to like elements and in which:
(2) FIG. **1** is an exploded, perspective view of a flow control device in accordance with one particular embodiment of the invention;
(3) FIG. **2** is an exploded view, taken from the side, of the flow control device of FIG. **1**;
(4) FIG. **3** is a top plan view of a flow control device in accordance with one particular embodiment of the invention;
(5) FIG. **3**A is a cut-away view of the flow control device of FIG. **3**, taken along the section lines A-A;
(6) FIG. **4**A is a side plan view of a portion of a flow control device in accordance with one particular embodiment of the invention;
(7) FIG. **4**B is a top plan view of the portion of the flow control device of FIG. **4**A;
(8) FIG. **5**A is a side plan view of a portion of a flow control device in accordance with one particular embodiment of the invention;
(9) FIG. **5**B is a top plan view of the portion of the flow control device of FIG. **5**A;
(10) FIG. **6**A is a side plan view of a portion of a flow control device in accordance with one particular embodiment of the invention;
(11) FIG. **6**B is a top plan view of the portion of the flow control device of FIG. **6**A;
(12) FIG. **7** is an exploded, perspective view of a flow control device in accordance with another particular embodiment of the invention;
(13) FIG. **8** is an exploded view, taken from the side, of the flow control device of FIG. **7**;
(14) FIG. **9** is a top plan view of a flow control device in accordance with one particular embodiment of the invention;
(15) FIG. **9**A is a cut-away view of the flow control device of FIG. **9**, taken along the section lines A′-A′;

(16) FIG. **10**A is a side plan view of a portion of a flow control device in accordance with one particular embodiment of the invention;
(17) FIG. **10**B is a top plan view of the portion of the flow control device of FIG. **10**A;
(18) FIG. **11**A is a side plan view of a portion of a flow control device in accordance with one particular embodiment of the invention;
(19) FIG. **11**B is a top plan view of the portion of the flow control device of FIG. **11**A;
(20) FIG. **12**A is a side plan view of a portion of a flow control device in accordance with one particular embodiment of the invention;
(21) FIG. **12**B is a top plan view of the portion of the flow control device of FIG. **12**A;
(22) FIG. **13** is an exploded, perspective view of a flow control device in accordance with a further particular embodiment of the invention;
(23) FIG. **14** is an exploded view, taken from the side, of the flow control device of FIG. **13**;
(24) FIG. **15** is an exploded, perspective view of a flow control device in accordance with still another particular embodiment of the invention;
(25) FIG. **16** is an exploded view, taken from the side, of the flow control device of FIG. **15**;
(26) FIG. **17** is a top plan view of a flow control device in accordance with one particular embodiment of the invention;
(27) FIG. **17**A is a cut-away view of the flow control device of FIG. **17**, taken along the section lines A″-A″;
(28) FIG. **18** is a top plan view of the portion of the flow control device in accordance with one particular embodiment of the invention;
(29) FIG. **19** is a perspective view of an infusion system with a pump, a syringe and a flow regulator in accordance with one particular embodiment of the present invention;
(30) FIG. **20** is a perspective view of a flow regulator according to the invention and its connectors to the remainder of the infusion system;
(31) FIG. **21** is a perspective view of a flow control device in accordance with still another particular embodiment of the present invention;
(32) FIG. **22** is a perspective view of an infusion pump that can be used in an infusion system in accordance with one particular embodiment of the invention; and
(33) FIG. **23** is a perspective view of an exemplary luer lock for use in one particular embodiment of an infusion system in accordance with the present invention.
(34) FIG. **24** is an exploded, perspective view of a flow control device in accordance with still another particular embodiment of the invention;

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

(35) Referring now to FIGS. **1-6**B, there is shown a variable flow control device **100** in accordance with one particular embodiment of the present invention. The flow control device **100** optimizes the delivery of fluids in conjunction with non-electric infusion pumps and gravity flow, so as to control the infusion of fluids for infusion therapy administration without the use of electronic infusion devices. The flow control device **100** is a variable flow regulator that can be provided as part of a complete infusion system or set. One example of one such complete infusion system or set is illustrated in FIG. **19**.
(36) In the present preferred embodiment, the flow control device **100** of the present invention is constructed with biocompatible materials. Preferably, flow control device **100** is designed with a geometry that is conducive to hand manipulation with sufficient gripping areas to avoid slippage and to facilitate rotation as a way to select a specific flow rate. The flow control device **100** is made with materials that allow for the infusion apparatus to be operated under gravity, as well as, pressurized at higher pressure ranges, such as from 5 PSI-40 PSI, as required by elastomeric and mechanical infusion devices designed to deliver fluids that require pressurized chambers in the 5-40 PSI range. The main flow rate variation is accomplished by adjusting the fluid path dimensions. Rotation of one half of the flow regulator component in one direction moves the fluid path exit

point along the channel and effectively changes total fluid path length and diameter such that fluid flow decreases or stops depending on the degree to which it is rotated. Rotation in the opposite direction moves the fluid path exit point towards the upstream fluid path entry point, changing effective internal fluid path length and diameter such that flow is increased.

(37) The flow control device **100** includes three primary elements: an inlet handle **110**; an outlet handle **130**; and a seal **120**, enclosed (i.e., "sandwiched" or sealingly engaged) between the inlet handle **110** and the outlet handle **130**. The outer surface of the inlet handle **110** includes a circumferential face or viewing portion **112** upon which a scale **114** showing selectable flow rates is imprinted. In the present embodiment, the inlet handle **110** and scale **114** provide a flow regulator dial that is rotatable relative to the outlet handle **130** to control the fluid flow through the device **100**.

(38) The inlet handle **110** includes a port **116** which serves as the fluid inlet to the device **100**. The port **116** includes a distal orifice **116***a* that allows fluid input to the port **116** to flow to the rest of the device **100**. The inlet handle **110** additionally includes a shaft **118** that includes a collar portion **118***a* that engages a snap fitting **138** of the outlet handle **130** to form a rotatable snap-fit coupling which holds the seal **120** in place between the inlet handle **110** and the outlet handle **130**. More particularly, the shaft **118** passes through a central hole **122** in the seal **120** and is entrapped in the snap fitting **138** of the outlet handle **130**, by its collar **118***a*.

(39) The outlet handle **130** includes an internal groove or "channel" **132** open at the top and of varying diameter, through which fluid passes to an orifice **132***a* at one end of the groove **132** connected to an outlet port **134**. The relative positions (i.e., overlap) of the inlet handle orifice **116***a* and the outlet handle groove **132** determines the flow rate. The outlet port **134** of the outlet handle **130** permits fluid to flow from the device **100** to tubing connected to a connector, preferably some form of universal connector, to allow connection to a patient.

(40) During assembly, the seal **120** is seated into an area of similar geometry to the seal **120** in the order to maintain the hole or orifice **124** through the seal **120** in alignment with the inlet handle orifice **116***a*. For example, as can be seen more particularly from FIGS. **5**B and **6**B, the lower portion of the inlet handle **110** includes a chamber or cavity **117**, sized and shaped to receive the seal **120** without permitting slippage. For example, the projections **126** on the seal **120** fit into mating recesses **117***a* in the cavity **117** to prevent the seal **120** from moving in the cavity **117**, and thus maintaining the orifice **116***a* in direct alignment with the orifice **124**, as shown more particularly in FIG. **3**A. This alignment permits fluid from the inlet handle **110** to flow through the seal **120** and into the groove **132** of the outlet handle. The seal orifice **124** is aligned with the orifice **116***a* and groove **132**, both during assembly and during flow. The seal **120** ensures that fluid is contained within the groove **132**, and that fluid input to the device **100** via the port **116** can only flow in a path defined by the orifices in each of the inlet handle **110** and seal **120** and groove **132** of the outlet handle **130**, to exit the device **100** through the port **134** of the outlet handle **130**.

(41) The inlet handle **110** and outlet handle **130** are preferably made of a material sufficiently robust to withstand the pressures of the intended use. In one particular embodiment of the invention, it is intended that the device **100** be used in a pressurized infusion system. Consequently, the material selected for inlet handle **110** and outlet handle **130** is preferably selected to operate under a wide range of pressures from 5-40 PSI (i.e., the device being operable for the entire range), making the device **100** compatible with pressurized devices. In one particular preferred embodiment, the material for the inlet and outlet handles **110**, **130** are selected to be polycarbonate or other materials of similar hardness coefficient. For example, the inlet and outlet handles can be made of a hard plastic to ensure precise sealing. The seal **120** is made of a soft plastic to provide a cushioned seal when placed between the inlet and outlet handles **110**, **130** and seals the device to prevent fluid leakage. The tolerance of the molds to produce the inlet and outlet handles should take into account that the inlet and outlet handles should be of sufficient tightness to avoid fluid leakage.

(42) Fluid viscosity, relative position of device, atmospheric pressure, ambient temperature and other factors affect actual flow rates. Calibration of flow rates and enhanced controllability are important clinical features. The flow control device of the present invention may be calibrated and delivered to the user with charts that correlate fluid viscosity with flow rate for various fluids under various conditions as part of the operating manual. The charts provided may include adjustment factors to account for and compensate for, among other factors that affect actual flow rates, fluid viscosity, relative position of device, atmospheric pressure, and/or ambient temperature and other factors.
(43) Conventional flow regulators are rated in ml/hr (milliliters per hour) and based on gravity flow for a low viscosity “Saline solution”. This way of labeling flow regulators is misleading and cannot be correlated to parametric variations. Instead, the device **100** has a numbering system that replaces the otherwise imprecise ml/hr indicators with numbers either from 1-6, 1-10 or 1-100% thus avoiding empirical discrepancies when different fluids are utilized. For example, in the present particular embodiment shown in FIGS. **1** and **2**, the scale **114** on the device **100** is labeled 1-6, which in the present example are not a numerical value in ml/hr. Each number on the scale **114** is aligned with a hash mark **114***a*, and the scale **114** additionally includes intervening hash marks **114***a* disposed halfway between the numbers, which hash marks **114** are alignable with an indicator or arrow **136** on the outlet handle **130**, for easy and understandable selection of the flow. More particularly, the inlet handle **110** and gasket or seal **120**, with the aligned orifices, **116***a*, **124**, rotate relative to the groove **132** of the outlet handle **130**, to align the orifices **116***a*, **124** with different portions of the groove **132**, thus controlling the flow between the inlet port **116** and the outlet port **134**.
(44) Referring now to FIGS. **7-12**B, there is shown a variable flow control device **200** in accordance with another particular embodiment of the present invention. The flow control device **200** is similar in many respects to that of FIGS. **1-6**B. More particularly, the flow control device **200** includes an inlet handle **210** including an inlet port **216**, orifice **216***a*, scale **214**, chamber or cavity **217** and shaft **218**, all of which operate similarly to the correspondingly named parts described in connection with FIGS. **1-6**B. The flow control device **200** additionally includes a seal **220** and an outlet handle **230**. However, instead of a single orifice, the seal **210** includes a plurality of orifices **222** alignable between the orifice **216***a* of the inlet handle **210** and a channel **236** containing an outlet orifice **232** connected to the outlet port **234** of the outlet handle **230**.
(45) However, the scale on the outlet handle **210** is mounted, in the present embodiment, on a dial **214** that can be rotated relative to the body of the inlet handle **210**. The chamber **217** containing the seal **220** forms the base of the dial **214**, so that, when received in the chamber **217**, the seal **220** is rotated when the dial **214** is rotated. Rotation of the dial **214** to a discretely marked position will align one of the orifices **222** (or no orifice, in the case of the “OFF” setting) with the inlet orifice **216***a* and with the channel **236** of the outlet handle **230**. The channel **236** is sized to receive fluid from any of the holes **222** and channel it to the outlet orifice **232** at the base of the channel **236**.
(46) In one particular embodiment of the invention, the scale on the dial **214** is operable between Off and 10 and the seal **220** includes 10 orifices **222**. Rotation of the dial **214** relative to the arrow or indicator **235** on the outlet handle **230** places a different orifice **222** between the inlet orifice **216***a* and the channel **236** containing the outlet orifice **232**. Each of the different orifices **222** are differently sized from one another to provide a correspondingly different flow through the seal **220**, and thus out the outlet port **234**. In the present example each orifice **222** is sized to provide a percentage of flow through the seal **220**. In the example shown, each of the 10 markings on the scale of the dial **214** represents 10% of the flow, such that aligning the number 1 on the dial **214** with the arrow **235** aligns an orifice **222** that permits fluid to flow at a flow rate of 10% of the total possible flow rate, between the inlet orifice **216***a* and the outlet orifice **232**. Similarly, selecting the hash mark next to the number 2 represents 20% of the total possible flow rate, while selecting the hash mark next to the number 10 represents 100% of the total possible flow rate. Although the

present example uses 10 discrete orifices **222** to provide flow rates changeable at 10% increments, this is not meant to be limiting, as more or fewer orifices **222** can be used. For example, if desired, 100 orifices **222** can be provided to permit the selection of a flow rate between 0 and 100% in 1% increments. Other numbers of orifices **222** can be used without deviating from the spirit of the invention.
(47) Additionally, the flow control device **200** is preferably made of the same materials, and for operation in the same pressure range, as the device **100**, described above. Additionally, the device **200** is assembled using a snap-fit coupling between a shaft **218**, having a collar **218***a*, and a snap fitting **238** on the outlet handle **230**, with the seal **220** disposed there between.
(48) Alternately, if desired, instead of a plurality of orifices **222** being provided on the seal **220**, the plurality of orifices can be provided on a face of the outlet handle, as shown more particularly in the embodiment of FIGS. **13** and **14**. Referring more particularly to FIGS. **13** and **14**, the seal can be the seal **120**, as described in connection with the embodiment of FIG. **1**. Similarly, the inlet handle can be the inlet handle **110** described in connection with the embodiment of FIG. **1**. Thus, the combination of inlet handle **110** and seal **120** would operate as described in connection with FIG. **1** (i.e., with the single orifice of the seal **120** fixedly aligned with the orifice **116***a* of the inlet handle **110**). However, instead of the outlet handle **130**, the device **300** includes the outlet handle **330**, the face of which includes a plurality of orifices **332**, each of a different size, as described in connection with the orifices **222** of FIG. **7**. The outlet handle **330** contains various internal “channels” connecting the orifices **332** to the outlet port **334**, through which the fluid travels. Rotation of the inlet handle **110** relative to the outlet handle **330** provides fluid from the inlet port **116**, via the orifices **116***a* (see, for example, FIG. **3**A) and **124**, to an aligned one of the orifices **332**. The diameter of the aligned one of the orifices **332** and/or the channel connecting it to the port **334**, define the rate of flow.
(49) As with the other embodiments, the numbers on the scale **114** can be aligned with the arrow or indicator to align the orifices **116***a*, **124** with a particular desired orifice **332** and provide the fluid to the outlet port **334** at the rate defined by the particular respective orifice **332**.
(50) Referring now to FIGS. **15-18**, there is shown a flow control device **400** in accordance with a further embodiment of the present invention. The device **400** includes an inlet handle **410** having a rotating dial **414**, an outlet handle **430** and a seal **420** including specialized groupings of orifices **422** defined for each possible discrete setting for the dial **414**. The flow control device **400**, i.e., the elements **410**, **420** and **430**, is preferably made of the same materials, and for operation in the same pressure range, as the corresponding elements of the device **100**, described above. Additionally, the device **400** is assembled using a snap-fit coupling between a shaft and snap fitting, with the seal **420** disposed therebetween, as was described in connection with the previous embodiments.
(51) Additionally, in the present embodiment, the rotation of the flow regulator dial **414** causes the flow from the inlet orifice or port **416** to be connected to the outlet handle **430**, via a particular group of orifices **422** in the seal **420**. More particularly, the seal **420** includes fifteen discrete positions, each of which has a unique group or combination **422** of orifices representing a specific binary number. In the present example, the inlet orifice **416***a* may be a single orifice, as shown, which is large enough to feed fluid to all of the orifices **422***a* in a particular group **422**. Alternately, if desired, the inlet **416** can be configured to have four orifices **416***a*, one for alignment with each of the four possible orifice locations in a group **422** on the seal **420**. Additionally, if desired, the orifice **416***a* can be in the form of a slot or groove (such as the groove **436** on the outlet handle), to ensure that fluid from the inlet port **216** will be provided to any open orifice in a group of orifices **422**. The number of orifices open between the inlet orifice **416***a*, and the outlet slot **436** and orifice **432**, as well as their sizes, define the flow rate for each particular position or setting of the dial **414**. As can be seen, with the use of four possible orifices **422***a* per group **422**, there are fifteen possible flow rate settings available to the flow rate device **400**, as shown in Table 1, here below.
(52) TABLE-US-00001 TABLE 1 ORIFICE POSITION No. ONE TWO THREE FOUR 1 1 0 0 0

2 0 1 0 0 3 1 1 0 0 4 0 0 1 0 5 1 0 1 0 6 0 1 1 0 7 1 1 1 0 8 0 0 0 1 9 1 0 0 1 10 0 1 0 1 11 1 1 0 1 12 0 0 1 1 13 1 0 1 1 14 0 1 1 1 15 1 1 1 1

(53) As shown more particularly in FIG. **18**, the size of each orifice **422***a* in a group **422** is, preferably, different from the size of every other orifice **422***a* in the group **422**, which provides the high resolution of fifteen unique possible dial positions. Additionally, the size of the holes should gradually increase. For example, in order to effectuate the fifteen unique settings of Table 1, in the present illustrative example, the orifice 2 must be larger than (i.e., have a greater flow through) the orifice 1. Similarly, orifice 3 must be larger than orifices 1 and 2, combined, and orifice 4 must be larger than orifices 1, 2 and 3 combined. In one particular embodiment of the invention, the second orifice is double the flow rate of the first orifice, the third orifice is double the flow rate of the second orifice and the fourth orifice is double the flow rate of the third orifice, and so on for the total number of orifices used. Note that that use of four orifices per group is not meant to be limiting, as more or fewer orifices **422***a* per group **422** may be used without departing from the scope of the invention. For example, in another particular embodiment of the invention (not shown), each group of orifices could have between 1 and 3 orifices, thus defining 8 unique flow rate settings, as defined by Table 2, herebelow.

(54) TABLE-US-00002 TABLE 2 ORIFICE POSITION No. ONE TWO THREE 1 1 0 0 2 0 1 0 3 1 1 0 4 0 0 1 5 1 0 1 6 0 1 1 7 0 1 1 8 1 1 1

(55) Additionally, if desired, in addition to, or instead of, the seal having groups of orifices, as described, the outlet handle, itself, may include a sequence of orifices, defined by Table 1, that permit fluid to flow through from seal into one such group of orifices. The relative position of the inlet handle and seal orifices with a particular group of orifices in the outlet handle determines the flow rate. One particular example of a flow rate device **500**, wherein the orifice groups **532** are on the outlet handle, instead of on the seal, is shown in FIG. **24**. More particularly, the inlet handle orifice is aligned with a slot **522** on the seal **520**, which can be aligned with each linear orifice group of **532** on the outlet handle **530** by rotating the inlet handle **510** relative to the outlet handle to align with a setting marked on the dial **514**. This changes the position of the slot **522** (and inlet orifice **516**) relative to the surface of the outlet handle **530**, and aligns the slot **522** with one of the particular orifice groups **532**. Channels in the outlet handle **530** direct the fluid from each orifice **532***a* of a group **532** to the outlet port (Not shown) of the device **500**.

(56) Referring now to FIGS. **19-20**, there is shown a pump **1** and a syringe **2** inserted in the pump **1**. A flow regulator **3** according to the invention is connected to the syringe **2**. The flow control device **3** can be any of the devices **100**, **200**, **300**, **400** discussed hereinabove. The pump **1** is designed with a pressure rating that is higher than elastomeric devices. The syringe **2** is a conventional, commercially available syringe. The flow regulator is provided with the necessary tubing and with commercially available universal female and male luer lock connectors. The female luer lock **4** is connected to the source of infusion and the male luer lock **8** is connected to the patient via a catheter or an additional extension infusion set.

(57) FIG. **22** shows the infusion pump **1** of FIG. **19** on a larger scale. The illustrated infusion pump **1** is the preferred embodiment in the context, but it will be understood that the flow regulator **3** according to the invention may also be used with other infusion pumps. The novel pump has label surface **10** for branding and the like. A rotating knob **11** or handle **11** may be used to initialize the pump. A viewing window **12** is provided for ascertaining that the syringe is properly inserted in the pump **1**.

(58) The following sequence may be performed by the user/patient in order to initialize the system and start the delivery of the infusion medicament: First, the regulator **3** must be set to zero in order to block any flow there through. Then the luer lock connector is attached to the filled syringe **2**. Next, the needle set is connected to the luer lock on the regulator **3**. Then the pump drive is opened by rotating the handle **11** counterclockwise until it stops. Then the syringe **2** is loaded and locked into the pump **2** by inserting the syringe plunger into the pump **1** and rotating the syringe 90° until

it “clicks” in place. Next, the user can verify that the syringe flange shows in the viewing window **12**, so as to confirm that the syringe is properly loaded. Now, the pump is ready for activation. It is activated by rotating the handle **11** clockwise. The system is primed by turning the regulator dial to position **5**, for instance, and, when the first drop of fluid comes out of the needle set, turning it back to the zero position. Now the infusion can be started as prescribed by the health provider. The user should thereby refer to the flow control guide. The delivery may be stopped by turning the regulator dial to the zero or off position. The syringe may be removed after the regulator has been set to zero and the handle **11** is rotated counterclockwise until the stop is reached.

(59) FIG. **23** is an enlarged view of a luer lock **4** for use in the novel infusion system. The luer lock is particularly easy to handle due to its ergonomic geometry.

(60) The entire flow control device **3** and connection system is illustrated in FIG. **19**. The female luer lock **4** is provided for conventional connection to the delivery end of the syringe **2**. A removable cap **13** is provided so as to protect the luer lock **4** during shipping and storage. A tubing section **6** leads from the luer lock **4** to the inlet side of the flow regulator **3**. The outlet side of the flow regulator **3** is connected via tubing **7** to a further connector **8** (here, a male luer lock), which is connected into the delivery IV tubing system **9**.

(61) The flow control device or regulator of the present invention is different from all other flow regulators in the market, in part, because all others are unable to withstand the considerable pressures supplied by the pump **1**, or other pumps with similar pressure ratings. Additionally, the flow control device of the present invention provides greater resolution for more accurately controlling the rate of fluid flow.

(62) As can be seen from the foregoing, flow control device **3** allows for the infusion pump to infuse a rate that is controllable by the user. Additional flow regulation control can be accomplished by incorporating one or more supplemental adjustable dial layers. For example, referring now to FIG. **21**, there is shown another embodiment of a flow rate control device **20**, wherein two dials **22**, **24** are used to improve precision in controlling flow rate. The first layer (**22**, **30**) provides a coarse control with methods described in connection with the foregoing embodiments discussed above. The second layer (**24**, **32**) provides a fine control, again with methods described in the foregoing embodiments discussed above. This is not meant to be limiting, as additional layers may be provided without departing from the scope of the invention.

(63) More particularly, as shown in FIG. **21**, the two layers are designed to provide a combination of coarse and fine control levels, thus greatly enhancing the actual flow rate controllability based on the selections/adjustments made on the first dial **22** and second dial **24**. Each of the dials **22**, **24** may work in accordance with the principles described herein. The second dial **24** may provide fine flow control, further refining the output controlled by the first dial **22**, which provides coarse flow control. In one particular embodiment, the second dial has a base diameter ranging between 0-10% of the first dial, which has a diameter ranging between 0-100% (coarse flow control). The combination of the first and the second dials **22**, **24** yields significantly enhanced control. Additional layers having a compounding effect on enhanced resolution, controllability and accuracy of the device **20** may be provided. Each layer may use the same or a different flow control mechanism as any other layer, as desired. For example, the coarse layer, including the dial **22** and outlet handle portion **30**, can make use of a variable diameter groove for controlling flow, such as described in connection with the device **100** FIG. **1**, while the second layer, including the dial **24** and the outlet handle portion **32** may make use of one of the other flow control mechanisms described in connection with the devices **200**, **300**, **400**, **500**. This is not meant to be limiting, as any combination of layers and any number of layers utilizing the variable flow control mechanisms described in connection with the devices **100**, **200**, **300**, **400**, **500** may be used in any of the layers without departing from the spirit of the present invention.

(64) As can be seen from the foregoing, the present invention implements features that improve flow control as well as safety resulting from such improved flow rate control when compared to the

performance of devices in the prior art. The invention solves important limitations inherent to non-electric infusion systems. Non electric infusions systems are generally controlled by certain small diameter tubing (rate set) that regulates the flow. This method presents limitations including inability to change flow rate without changing the rate set, incorrect flow rate labeling due to the varying viscosities of fluids administered, and undesired flow rates due to device design limitations, patient conditions and environmental factors.
(65) The flow rate control devices described hereinabove can be used in connection with gravity infusion devices and/or pressurized infusion devices, including constant or semi-constant force infusion devices. Additionally, if desired, a pressure sensor may be provided the output of which can be used to by self-adjusting mechanism or circuit to automatically adjust the flow rate of the flow rate device to reduce flow changes relative to pressure variations, patient conditions and other therapy factors.
(66) The present disclosure is provided to allow practice of the invention, after the expiration of any patent granted hereon, by those skilled in the art without undue experimentation, and includes the best mode presently contemplated and the presently preferred embodiment. Nothing in this disclosure is to be taken to limit the scope of the invention, which is susceptible to numerous alterations, equivalents and substitutions without departing from the scope and spirit of the invention.

Claims

1. A flow rate device for fluid control along a flow path between an infusion device and a patient comprising: an inlet handle comprising an inlet port configured to receive fluid originating from the infusion device at a pressure within a range of 5 pounds per square inch (PSI) to 40 PSI; an outlet handle comprising an outlet port configured to deliver fluid to the patient, the inlet handle being rotatable relative to the outlet handle; and a fluid path disposed between the inlet handle and the outlet handle, the fluid path having at least one manually adjustable dimension which, when adjusted due to rotation of the inlet handle relative to the outlet handle during infusion, changes a flow rate of fluid exiting the outlet port, the at least one manually adjustable dimension including at least one of a width and a length of the fluid path.
2. The flow rate device of claim 1, wherein the inlet handle and the outlet handle are composed of materials that can withstand pressures induced by the infusion device of from 5 PSI to 40 PSI.
3. The flow rate device of claim 1 further comprising: a seal between and rotatable relative to one of the inlet handle or the outlet handle and fixed relative to the other of the inlet handle or the outlet handle.
4. The flow rate device of claim 1, wherein the infusion device is a non-electric infusion pump.
5. The flow rate device of claim 4, wherein the infusion device dispenses fluid at a constant force.
6. The flow rate device of claim 1, wherein the infusion device is coupled to the flow rate device by way of one or more tubing sets.
7. The flow rate device of claim 1, wherein one or more tubing sets are coupled to the outlet port to cause the fluid to be delivered to the patient.
8. The flow rate device of claim 1, wherein the outlet handle comprises a channel of varying diameter, wherein a position of the outlet handle relative to the inlet handle changes a position at which the fluid enters the channel thereby affecting the flow rate of fluid exiting the outlet port.
9. The flow rate device of claim 8 further comprising: a seal between the inlet handle and the outlet handle having a plurality of orifices for fluid flow therethrough, each of the orifices being selectively alignable with the inlet port and the channel.
10. The flow rate device of claim 8 further comprising: a seal between the inlet handle and the outlet handle consisting of a single orifice for fluid flow therethrough, the orifice being selectively alignable with the inlet port and the channel.

11. A flow rate device for fluid control along a flow path between an infusion device and a patient comprising: an inlet handle comprising an inlet port configured to receive fluid originating from the infusion device at a pressure within a range of 5 pounds per square inch (PSI) to 40 PSI; an outlet handle comprising an outlet port configured to deliver fluid to the patient, the inlet handle being rotatable relative to the outlet handle; and a fluid path disposed between the inlet handle and the outlet handle, the fluid path having a manually adjustable fluid path length which, when adjusted due to rotation of the inlet handle relative to the outlet handle during infusion, changes a flow rate of fluid exiting the outlet port.
12. The flow rate device of claim 11, wherein the inlet handle and the outlet handle are composed of materials that can withstand pressures induced by the infusion device of from 5 PSI to 40 PSI.
13. The flow rate device of claim 11 further comprising: a seal between and rotatable relative to one of the inlet handle or the outlet handle and fixed relative to the other of the inlet handle or the outlet handle.
14. The flow rate device of claim 11, wherein the infusion device is a non-electric infusion pump.
15. The flow rate device of claim 14, wherein the infusion device dispenses fluid at a constant force.
16. The flow rate device of claim 11, wherein the infusion device is coupled to the flow rate device by way of one or more tubing sets.
17. The flow rate device of claim 11, wherein one or more tubing sets are coupled to the outlet port to cause the fluid to be delivered to the patient.
18. The flow rate device of claim 11, wherein the outlet handle comprises a channel of varying diameter, wherein a position of the outlet handle relative to the inlet handle changes a position at which the fluid enters the channel thereby affecting the flow rate of fluid exiting the outlet port.
19. The flow rate device of claim 18 further comprising: a seal between the inlet handle and the outlet handle having a plurality of orifices for fluid flow therethrough, each of the orifices being selectively alignable with the inlet port and the channel.
20. The flow rate device of claim 18 further comprising: a seal between the inlet handle and the outlet handle consisting of a single orifice for fluid flow therethrough, the orifice being selectively alignable with the inlet port and the channel.
21. A flow rate device for fluid control along a flow path between an infusion device and a patient comprising: an inlet handle comprising an inlet port configured to receive fluid originating from the infusion device at a pressure within a range of 5 pounds per square inch (PSI) to 40 PSI; an outlet handle comprising an outlet port configured to deliver fluid to the patient, the inlet handle being rotatable relative to the outlet handle; and a fluid path disposed between the inlet handle and the outlet handle, the fluid path having a manually adjustable width which, when adjusted due to rotation of the inlet handle relative to the outlet handle during infusion, changes a flow rate of fluid exiting the outlet port.